MILITARY SPECIFICATION

TESTER, AIRCRAFT PERSONNEL SAFETY EQUIPMENT

1. SCOPE

1.1 This specification covers a tester for aircraft safety belts, harnesses, and control cables (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 --Boxes, Wood, Cleated-Plywood.
PPP-B-621 --Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 --Box, Fiberboard.
PPP-B-640 --Boxes, Fiberboard, Corrugated, Triple, Wall.
PPP-C-843 --Cushioning Material, Cellulosic.

MILITARY

MIL-H-5606 --Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-H-6083 --Hydraulic Fluid, Petroleum Base, Preservative.

STANDARDS

MILITARY

MIL-STD-129 --Marking for Shipment and Storage.
MIL-STD-130 --Identification Marking of U.S. Military Property.
MIL-STD-143 --Specifications and Standards Order of Precedence for the Selection of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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2.1.2 DRAWINGS

TRANSPORTATION MATERIEL COMMAND

4920-TE-017 --Tester Installation--Cable and Harness.
4920-TE-018 --Hydraulic Schematic--Tester, Cable and Harness.
4920-TE-019 --Frame--Tester, Cable and Harness.
4920-TE-020 --Frame Fittings--Tester, Cable and Harness.
4920-TE-021 --Safety Guard Assembly--Tester, Cable and Harness.
4920-TE-022 --MA-1 Dummy Assy.--Tester, Cable and Harness.
4920-TE-023 --Retainers--Tester, Cable and Harness.
4920-TE-024 --Cable Adapters--Tester, Cable and Harness.
4920-TE-025 --Assy Attachment--Tester, Cable and Harness.
4920-TE-026 --Parts Case Assy--Tester, Cable and Harness.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. REQUIREMENTS

3.1 Safety equipment tester. The safety equipment tester shall be a hydraulic tension machine consisting of frame, hydraulic system, and appropriate adapters. The tester shall be constructed in accordance with Drawings 4920-TE-017 through 4920-TE-026.

3.1.1 Preproduction sample. If specified (see 6.2), a preproduction model tester shall be submitted for testing as specified in 4.2.1. A preproduction sample is defined as a production line sample which the manufacturer will subsequently produce in fulfillment of the contract or order. Approval of the preproduction sample by the procuring activity shall not relieve the contractor of his responsibility to supply testers conforming to this specification. Any change of deviation from the approved preproduction sample shall be subject to the approval of the contracting officer.

3.2 Materials. Materials shall be as specified herein and on the applicable drawings. When specificatios and standards are not designated, selection shall be in accordance with Standard MIL-STD-143.

3.3 Components. Tester shall consist of frame, hydraulic system, and adapters.

3.3.1 Frame. The frame shall be constructed from sections of aluminum tubes which can be extended to test cables up to 96 inches long. The tubes shall be provided with brackets for supporting the hydraulic system and for bench mounting of the tester. A safety screen shall be provided for operator protection.

3.3.2 Hydraulic system. The hydraulic system shall consist of a two-way cylinder which is powered by a hand-operated pump. Selection of the direction of motion of the cylinder shall be by a four-way valve. The test load shall be

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determined by a pressure gage which is graduated in pounds per square inch (psi) and pounds tension. An adjustable relief valve shall maintain a safe hydraulic pressure limit.

3.3.3 Adapters. The adapters, and a wooden carrying case for storage of the adapters, shall be as specified on the drawings in 2.1.

3.4 Performance.

3.4.1 Hydraulic system. Hydraulic system components shall have a minimum operating pressure rating of 3,000 psi. The system shall be capable of maintaining this pressure without leakage at fittings or sels.

3.4.2 Structure. Tester frame, in the fully extended condition, shall be capable of withstanding a 6,300-pound load without permanent deformation of any part of the structure. When tested as specified in 4.4.2, the deflection of the main frame from the flat surface shall not exceed 5 inches.

3.4.3 Accuracy. The tester shall indicate the actual applied test load within an accuracy of 0.5-percent of the full scale reading (see 4.4.3).

3.5 Painting and finishing. Painting and finishing shall be as specified on the applicable drawings (see 2.1).

3.6 Instruction books. Instruction books shall be furnished as specified in the contract or order (see 6.2).

3.7 Identification of product. Tester shall be identified in accordance with Standard MIL-STD-130.

3.8 Workmanship. Workmanship shall be of a quality that will produce a tester assuring the required accuracy and durability. The tester shall be clean and free from scratches, dents, pits, rust, sharp edges, lose connections; omitted or improper machine or process operations; or other defects which would make the item unsuitable for the intended use. External edges of the carrying case, as well as of the tester, shall be rounded and beveled to prevent injury during handling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. Examinations and tests shall be classified as (1) preproduction and (2) conformance.

4.2.1 Preproduction. The preproduction sample of the tester (see 3.1.1) shall be subjected to all the examinations and tests specified herein.

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4.2.2 Conformance. Each tester shall be subjected to the following:

(a) Examination of product (see 4.3).

(b) System tests (see 4.4).

4.3 Examination. Each tester shall be examined for proper assembly of the hydraulic system and the frame and for the presence of all adapters and parts.

4.4 Tests.

4.4.1 Hydraulic system. In the assembled tester, the hydraulic system shall be filled to the proper level with hydraulic fluid conforming to Specification MIL-H-5606. With the relief valve set at maximum pressure, the ram bottomed at the end of its tension stroke, 3,000 psi shall be applied by using the hand pump. Pressure loss shall not exceed 100 psi during a 5-minute period. There shall be no evidence of leakage at fittings (see 3.4.1).

4.4.2 Structure. The assembled tester, with the sliding bracket at the 96-inch position, shall be placed on a flat surface, the mounting brackets unattached to the surface. A calibrated tension dynamometer shall be installed between the ram and the sliding bracket, with suitable cable and adapters. When the 6,300-pound load is applied as specified in 3.4.2, the deflection of the main frame from the flat surface shall not exceed 5 inches. The load shall then be released, the tester disassembled, and the structure examined for permanent deflection. Failure due to defective material or workmanship shall be cause for rejection.

4.4.3 Accuracy. Gage shall be removed and calibrated over entire range at 10-percent intervals with a deadweight tester. Gage shall be accurate within .05-percent of the full scale (15 psi). Tester shall then be reassembled to the most convenient length and mounted to a bench or other flat surface. the calibrated tension dynamometer shall be reinstalled and loads of 500, 1,000, 1,500, 2,000, 3,000, 4,000, 5,000 and 5,500 pounds applied. The gage readings shall be accurate within 0.5-percent of the full scale (75 pounds). Relief valve shall be set to release pressure at 5,500 pounds.

4.5 Inspection of preparation for delivery. The preservation and packaging, packing, and marking of the tester shall be inspected to determine conformance with section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or level C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Hydraulic system. The hydraulic system shall be flushed with fluid conforming to Specification MIL-H-6083, and then drained. The cylinder rod shall be set in the retracted position, and the control valve closed.

5.1.1.2 Pressure gage. Pressure gage shall be removed, and gage port and snubber plugged or capped. The gage shall be cushioned with material conforming to Specificaiton PPP-C-843, and packaged in a fiberboard box conforming to Specification PPP-B-636.

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5.1.1.3 Main frame. The tubes comprising the main frame of the tester shall be disassembled, and mounting brackets and connectors removed. Retainer pins shall be stored in the adapter case.

5.1.1.4 Safety screen. The sections of the safety screen shall be nested, and attaching hardware bagged and stored in the adapter case.

5.1.1.5 Adapter case. Adapter case shall be securely latched to prevent opening during shipment and handling.

5.1.2 Level C. Testers shall be preserved and packaged in a manner to prevent deterioration and damage during shipment and handling from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, level B, or level C, as specified (see 6.2).

5.2.1 Level A. Testers preserved and packaged as specified in 5.1.1 shall be individually packed in cleated plywood boxes conforming to type II of Specification PPP-B-601, or in nailed wood boxes conforming to class 2 of Specification PPP-B-621, or in triple wall fiberboard boxes conforming to class 2 of Specification PPP-B-640. Contents of shipping containers shall be blocked and braced, or anchored as necessary to prevent movement. Sharp edges of components shall be padded or cushioned as necessary to prevent damage to any parts of the unit. Containers shall be closed and strapped in accordance with the applicable box specification, or appendix thereto.

5.2.2 Level B. Testers shall be packed as specified in 5.2.1, except that containers shall be of the type or class required for domestic use.

5.2.3 Level C. Testers shall be packed in a manner to assure acceptance by common carrier for shipment at the lowest applicable rate, and sfe delivery at destination. Containers shall conform to the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Safety equipment tester is intended for use in testing the condition of the material in aircraft safety belts, harnesses, and control cables.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether preproduction sample is required for testing (see 3.1.1).

(c) Whether instruction books are required, and if so, the quantity and the address to which they should be sent (see 3.6).

(d) Level of preparation for delivery required (see 5.1 and 5.2).

(e) Whether special marking is required (see 5.3).

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5. PROBLEM AREAS

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